R. Weizmann KIENDREBEOGO, PhD

Postdoctoral Researcher · IRFU / CEA Paris-Saclay, Gif-sur-Yvette, France

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Research Interests

Multi-messenger Astronomy · Gravitational Waves · Time-Domain Astronomy · Kilonovae and Compact Binary Mergers · Observation Planning and Scheduling · Bayesian Inference · Machine Learning and Deep Learning for Astronomy

Education

PhD in Planetary and Universe Sciences, Université Côte d'Azur, Nice, France — September 2024

M.Sc. (Master II) in Astrophysics & Materials — December 2020
Université Joseph Ki-Zerbo, Ouagadougou, Burkina Faso

B.Sc. in Fundamental Physics — October 2017
Université Nazi Boni, Bobo-Dioulasso, Burkina Faso

Two-year university degree in Mathematics, Physics & Computer Science (MPI) — 2016
Université Nazi Boni, Bobo-Dioulasso, Burkina Faso

Appointments

Postdoctoral Researcher, IRFU / CEA Paris-Saclay, France — January 2025 - present

Full-Time Physics and Chemistry Teacher (High School) — 2018 - 2021
Complexe Scolaire Sainte Marie Eugénie, Ouagadougou, Burkina Faso

Publications

*Author name in **bold**. Full list on **Google Scholar***

Selected Publications

1. **Kiendrebeogo** et al. (2026), "NMMA–Astro-COLIBRI: An Automated Light-Curve Classification Service for Supernovae in the Multi-Survey Era." *Submitted to The Astrophysical Journal Supplement Series*. https://doi.org/10.48550/arXiv.2608.17568
2. Johnson, Sravan, **Kiendrebeogo**, Coughlin, Davis, Toivonen, Jegou du Laz, Erb, Ahumada, Barna, Helou, Smith, Rusholme, Laher and Mahabal (2026), "Low-latency Forecasts of Kilonova Light Curves for Rubin and ZTF." *Publications of the Astronomical Society of the Pacific* **138**, 014103. https://doi.org/10.1088/1538-3873/ae2e87
3. **Kiendrebeogo** et al. (2025), "Application of Nonlinear Noise Regression in the Virgo Detector." *Physical Review D* **112**, 083053. https://doi.org/10.1103/hwnc-r7yy
4. Singer, Criswell, Leggio, **Kiendrebeogo**, Coughlin, Earnshaw, Gezari, Grefenstette, Harrison, Kasliwal, Morris, Tollerud and Cenko (2025), "Optimal Follow-up of Gravitational-wave Events with the UltraViolet EXplorer (UVEX)." *Publications of the Astronomical Society of the Pacific* **137**, 074501. https://doi.org/10.1088/1538-3873/adcfc6
5. Ahumada, . . . , **Kiendrebeogo**, . . . et al. (2024), "Searching for Gravitational Wave Optical Counterparts with the Zwicky Transient Facility: Summary of O4a." *Publications of the Astronomical Society of the Pacific* **136**, 114201. https://doi.org/10.1088/1538-3873/ad8265
6. **Kiendrebeogo** et al. (2023), "Updated Observing Scenarios and Multimessenger Implications for the International Gravitational-wave Networks O4 and O5." *The Astrophysical Journal* **958**, 158. https://doi.org/10.3847/1538-4357/acfcb1
7. Pang, Dietrich, Coughlin, Bulla, Tews, Almualla, Barna, **Kiendrebeogo**, Kunert, Mansingh, Reed, Sravan, Toivonen, Antier, Vanden Berg, Heinzel, Nedora, Salehi, Sharma, Somasundaram and Van Den Broeck (2023), "An Updated Nuclear-Physics and Multi-Messenger Astrophysics Framework for Binary Neutron Star Mergers." *Nature Communications* **14**, 8352. https://doi.org/10.1038/

s41467-023-43932-6

8. Coughlin, . . . , **Kiendrebeogo**, . . . et al. (2023), "A Data Science Platform to Enable Time-Domain Astronomy." *The Astrophysical Journal Supplement Series* **267**, 31. https://doi.org/10.3847/1538-4365/acdee1
9. Kunert, Antier, Nedora, Bulla, Pang, Anand, Coughlin, Tews, Barnes, Hussenot-Desenonges, Healy, Jegou du Laz, Pilloix, **Kiendrebeogo** and Dietrich (2023), "Bayesian Model Selection for GRB 211211A through Multiwavelength Analyses." *Monthly Notices of the Royal Astronomical Society* **527**, 3900–3911. https://doi.org/10.1093/mnras/stad3463
10. Ashkar, Seglar-Arroyo, Schüssler, **Kiendrebeogo** and de Bony de Lavergne (2025), "tilepy: Smart Scheduling for Multi-Messenger Astronomy from Earth to Orbit." Proceedings of the 39th International Cosmic Ray Conference (ICRC2025). https://doi.org/10.48550/arXiv.2508.07824

Selected Open-Source Contributions

1. `NMMA-Astro-COLIBRI`: https://nmma.live/ · https://github.com/astro-transients/nmma-astrocolibri
2. `Astro-COLIBRI`: https://astro-colibri.com
3. `SkyPortal`: https://github.com/skyportal/skyportal
4. `NMMA`: https://github.com/nuclear-multimessenger-astronomy/nmma
5. `M4OPT`: https://github.com/m4opt/m4opt
6. `tilepy`: https://github.com/astro-transients/tilepy
7. `GWEMOPT`: https://github.com/skyportal/gwemopt
8. `ligo.skymap`: https://lscsoft.docs.ligo.org/ligo.skymap/
9. `observing-scenarios-simulations`: https://github.com/lpsinger/observing-scenarios-simulations
10. `DeepClean / Virgo-DeepClean`: https://github.com/weizmannk/Virgo-DeepClean
11. `cbc-population-distributions`: https://cbc-population-distributions.readthedocs.io/en/latest/
12. `EarthOrbitPlan`: https://earthorbitplan.readthedocs.io/en/latest/

Technical Skills

Programming: Python (advanced), C/C++, Bash, Dart / Flutter (Astro-COLIBRI mobile app), LaTeX

Scientific computing: NumPy, SciPy, pandas, Astropy, Matplotlib, CuPy (GPU-accelerated arrays), HEALPix / ligo.skymap

Machine learning: PyTorch, TensorFlow, scikit-learn, Deep learning for time series (CNN)

Bayesian inference: bilby, NMMA, Nested sampling, MCMC, Population inference

Software engineering: Git (GitHub, GitLab), Docker, pytest, PyPI/conda, CI/CD, REST APIs, Technical documentation (Sphinx, MkDocs, Read the Docs)

High-performance computing: HTCondor and SLURM clusters, Large-scale simulation campaigns (>20M injections)

Languages

French (native) · English (fluent, working language) · Mooré (native)